

Appendix E

Common data models

In this appendix you will learn:

- More about building logical data models.
 - About common logical data models.
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In this appendix, we introduce some common data models that you may find useful. In fact, it has been estimated that one-third of a data model consists of common constructs that are applicable to most companies and the remaining two-thirds are either industry-specific or company-specific. Thus, most data modeling work is re-creating constructs that have already been produced many times before in other companies.

The two main aims of this appendix therefore are to provide you with:

- (1) additional knowledge of building data models;
- (2) data model templates that you may find useful in your business. The models featured here may not represent your company exactly, but they may provide a starting point from which you can develop a more suitable model that matches your company's specific requirements.

We provide models for the following common business areas:

- Customer order entry
- Inventory control
- Asset management
- Project management

- Course management
- Human resource management
- Payroll management.

We also provide the following data models which are less common but may still be useful from both a business perspective and a learning perspective:

- Vehicle rentals
- Student accommodation
- Client transportation
- Publisher printing
- County library
- Real estate rentals
- Travel agent
- Student results.

In each case, we provide a short description of the requirements, and show an example of a typical logical data model and the mapping of the model to a set of tables. We assume that you are familiar with the modeling notation used throughout this book. If you are not, look at Chapter 7 on ER modeling, which introduces the main concepts and notations we use in this appendix. You will also find a summary of the database design methodology in Appendix B.

E.1 Customer order entry

A company wishes to create a database for its order entry activities. A customer can place one or more orders, with each order for one or more products. Each order gives rise to one invoice, which can be paid by a number of methods, such as check, credit card, or cash. The name of the employee who initially processes the customer order is recorded.

An employee in the Shipping Department is responsible for packaging the order and sending it to the customer. If an ordered product is not in stock, Shipping send out what is in stock, so more than one shipment may be required to fulfill the order. The logical data model is shown in Figure E.1 and the associated tables in Figure E.2.

Figure E.1

Logical data model for customer order entry.

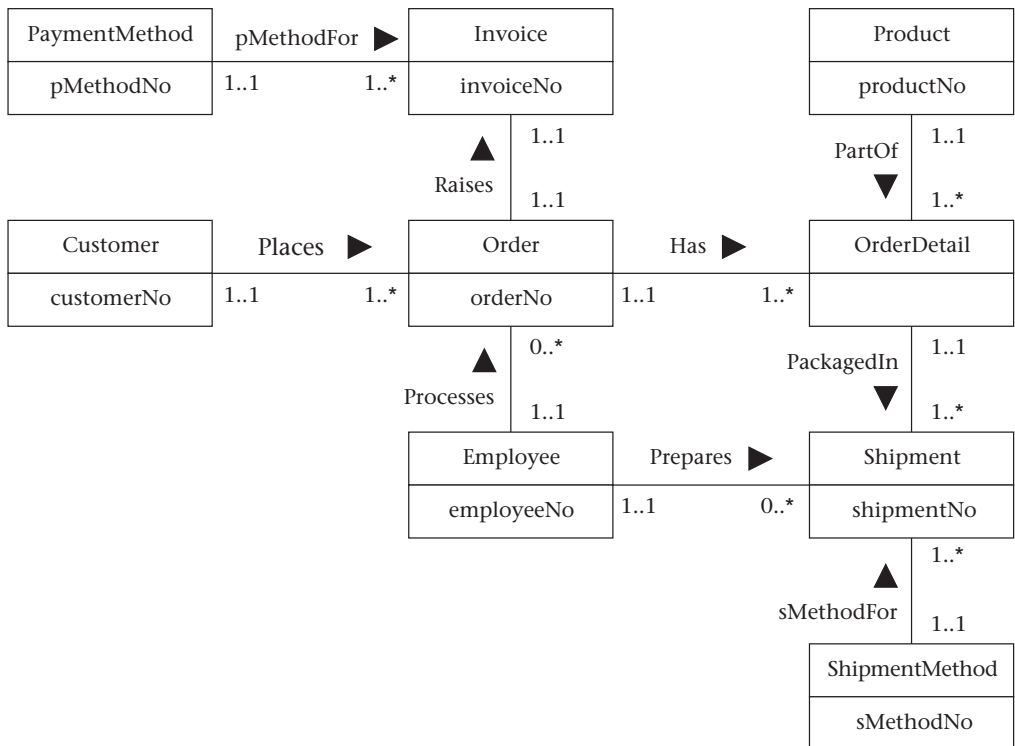


Figure E.2

Tables for customer order entry.

Customer	(<u>customerNo</u> , customerName, customerStreet, customerCity, customerState, customerZipCode, custTelNo, custFaxNo, DOB, maritalStatus, creditRating) Primary Key customerNo Alternate Key custTelNo Alternate Key custFaxNo
Employee	(<u>employeeNo</u> , title, firstName, middleName, lastName, address, workTelExt, homeTelNo, empEmailAddress, socialSecurityNumber, DOB, position, sex, salary, dateStarted) Primary Key employeeNo Alternate Key socialSecurityNumber
Invoice	(<u>invoiceNo</u> , dateRaised, datePaid, creditCardNo, holdersName, expiryDate, orderNo, pMethodNo) Primary Key invoiceNo Foreign Key orderNo references Order(orderNo) Foreign Key pMethodNo references PaymentMethod(pMethodNo)
Order	(<u>orderNo</u> , orderDate, billingStreet, billingCity, billingState, billingZipCode, promisedDate, status, customerNo, employeeNo) Primary Key orderNo Foreign Key customerNo references Customer(customerNo) Foreign Key employeeNo references Employee(employeeNo)
OrderDetail	(<u>orderNo</u> , <u>productNo</u> , quantityOrdered) Primary Key orderNo, productNo Foreign Key orderNo references Order(orderNo) Foreign Key productNo references Product(productNo)
PaymentMethod	(<u>pMethodNo</u> , paymentMethod) Primary Key pMethodNo
Product	(<u>productNo</u> , productName, serialNo, unitPrice, quantityOnHand, reorderLevel, reorderQuantity, reorderLeadTime) Primary Key productNo Alternate Key serialNo
Shipment	(<u>shipmentNo</u> , quantity, shipmentDate, completeStatus, orderNo, productNo, employeeNo, sMethodNo) Primary Key shipmentNo Foreign Key orderNo, productNo references OrderDetail(orderNo, productNo) Foreign Key employeeNo references Employee(employeeNo) Foreign Key sMethodNo references ShipmentMethod(sMethodNo)
ShipmentMethod	(<u>sMethodNo</u> , shipmentMethod) Primary Key sMethodNo

E.2 Inventory control

A company wishes to create a database to control its inventory, which consists of a number of products divided into a number of categories, such as clothing, food, and stationery. An employee raises a purchase order when a product has to be reordered from the supplier. The tracking records supplies received, units sold, and any wastage. The logical data model is shown in Figure E.3 and the associated tables in Figure E.4.

Figure E.3

Logical data model for inventory control.

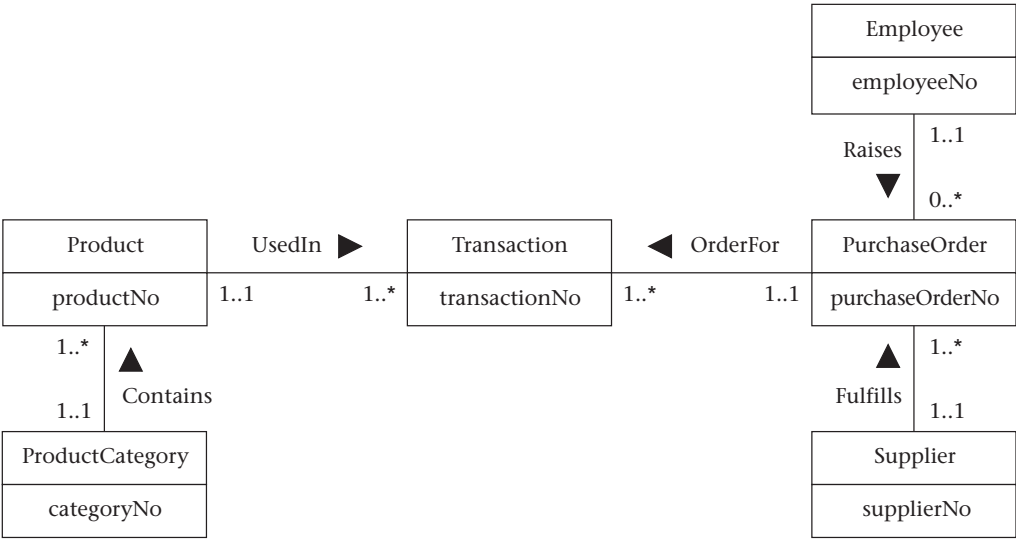


Figure E.4

Tables for inventory control.

Employee	As defined in Section D.1.2
Product	(<u>productNo</u> , productName, serialNo, unitPrice, quantityOnHand, reorderLevel, reorderQuantity, reorderLeadTime, categoryNo) Primary Key productNo Alternate Key serialNo Foreign Key categoryNo references ProductCategory(categoryNo)
ProductCategory	(<u>categoryNo</u> , categoryDescription) Primary Key categoryNo
PurchaseOrder	(<u>purchaseOrderNo</u> , purchaseOrderDescription, orderDate, dateRequired, shippedDate, freightCharge, supplierNo, employeeNo) Primary Key purchaseOrderNo Foreign Key supplierNo references Supplier(supplierNo) Foreign Key employeeNo references Employee(employeeNo)
Supplier	(<u>supplierNo</u> , supplierName, supplierStreet, supplierCity, supplierState, supplierZipCode, suppTelNo, suppFaxNo, suppEmailAddress, suppWebAddress, contactName, contactTelNo, contactFaxNo, contactEmailAddress, paymentTerms) Primary Key supplierNo Alternate Key supplierName Alternate Key suppTelNo Alternate Key suppFaxNo
Transaction	(<u>transactionNo</u> , transactionDate, transactionDescription, unitPrice, unitsOrdered, unitsReceived, unitsSold, unitsWastage, productNo, purchaseOrderNo) Primary Key transactionNo Foreign Key productNo references Product(productNo) Foreign Key purchaseOrderNo references PurchaseOrder(purchaseOrderNo)

E.3 Asset management

A company wishes to create a database to monitor each of its assets (such as PCs, printers, cars, desks, chairs). Assets are divided into a number of categories, such as computers and furniture. An asset is assigned to an employee. On a regular basis, an employee in the Finance Department checks each asset to determine its current market value, and records the date and the current value of the asset. As a result of the assessment, the company may decide to sell the asset. Also on a regular basis, maintenance is carried out on each asset. In some cases, the maintenance is carried out by an employee, but in others, the asset has to be sent to an external company for maintenance. The logical data model is shown in Figure E.5 and the associated tables in Figure E.6.

Figure E.5

Logical data model for asset management.

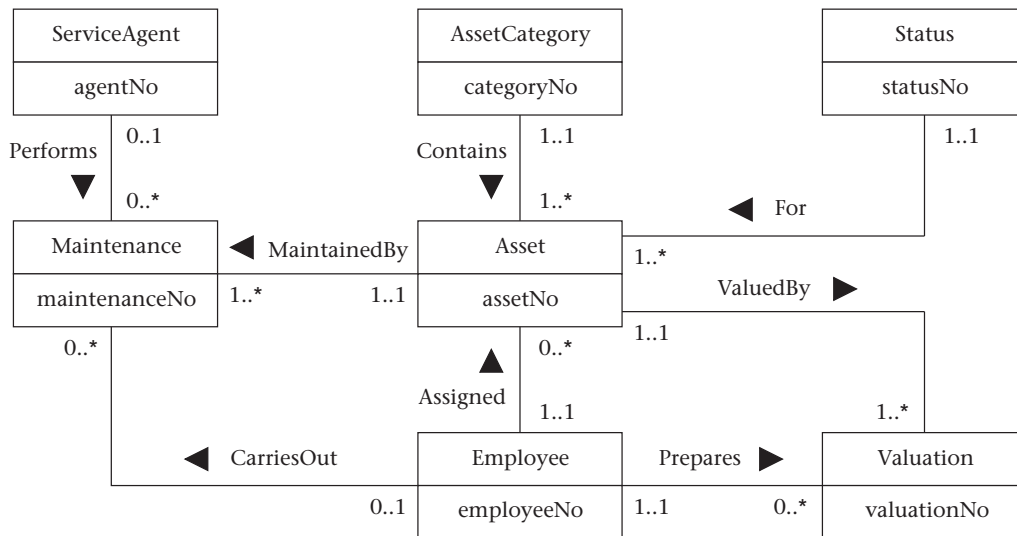


Figure E.6

Tables for asset management.

Employee	As defined in Section D.1.2
Asset	(<u>assetNo</u> , assetDescription, serialNo, dateAcquired, purchasePrice, currentValue, dateSold, nextMaintenanceDate, employeeNo, assetCategoryNo, statusNo) Primary Key assetNo Alternate Key serialNo Foreign Key employeeNo references Employee(employeeNo) Foreign Key assetCategoryNo references AssetCategory(assetCategoryNo) Foreign Key statusNo references Status(statusNo)
AssetCategory	(<u>assetCategoryNo</u> , assetCategoryDescription) Primary Key assetCategoryNo
Maintenance	(<u>maintenanceNo</u> , maintenanceDate, maintenanceDescription, maintenanceCost, assetNo, employeeNo, agentNo) Primary Key maintenanceNo Foreign Key assetNo references Asset(assetNo) Foreign Key employeeNo references Employee(employeeNo) Foreign Key agentNo references ServiceAgent(agentNo)
ServiceAgent	(<u>agentNo</u> , agentName, agentStreet, agentCity, agentState, agentZipCode, agentTelNo, agentFaxNo, agentEmailAddress, agentWebAddress, contactName, contactTelNo, contactFaxNo, contactEmailAddress) Primary Key agentNo Alternate Key agentName Alternate Key agentTelNo Alternate Key agentFaxNo
Status	(<u>statusNo</u> , statusDescription) Primary Key statusNo
Valuation	(<u>valuationNo</u> , valuationDate, valuationPrice, assetNo, employeeNo) Primary Key valuationNo Foreign Key assetNo references Asset(assetNo) Foreign Key employeeNo references Employee(employeeNo)

E.4 Project management

A consultancy company wishes to create a database to help manage its projects. Each project is for a specific client and has a nominated project manager. The project is divided into a number of work packages and employees bill their time and expenses against a work package. Each employee has a specific role, which defines the charging rate for the client. Over time, an employee can work on several work packages associated with the same project. In addition, most, but not all, work packages have a number of associated documents as deliverables, each of which may be written by more than one employee. The logical data model is shown in Figure E.7 and the associated tables in Figure E.8.

Figure E.7

Logical data model for project management.

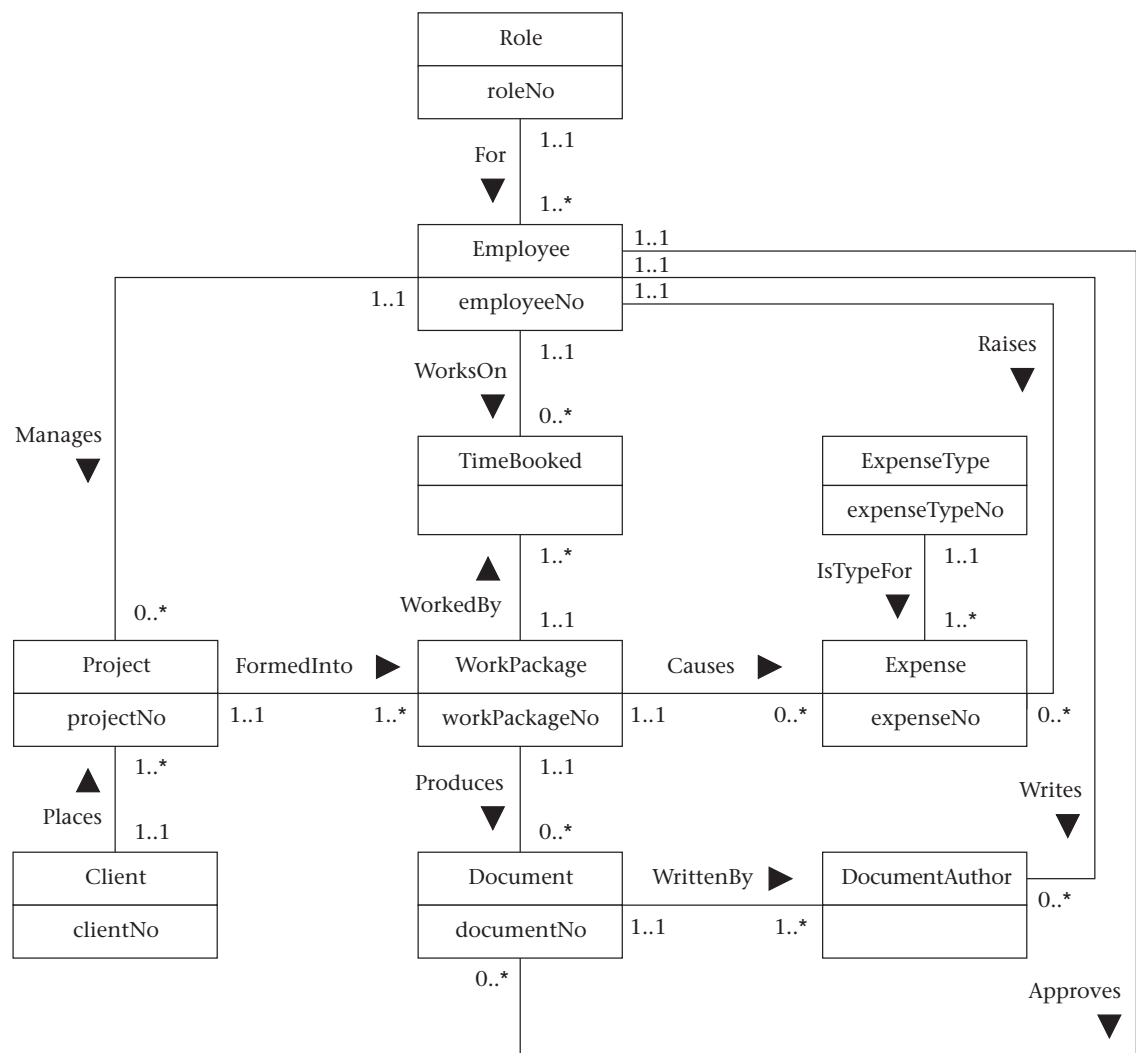


Figure E.8

Tables for project management.

Client	(<u>clientNo</u> , clientName, clientStreet, clientCity, clientState, clientZipCode, clientTelNo, clientFaxNo, clientWebAddress, contactName, contactTelNo, contactFaxNo, contactEmailAddress) Primary Key clientNo Alternate Key clientName Alternate Key clientTelNo Alternate Key clientFaxNo
Document	(<u>documentNo</u> , documentTitle, documentDate, versionNo, workPackageNo, approvedByEmployeeNo) Primary Key documentNo Foreign Key workPackageNo references WorkPackage(workPackageNo) Foreign Key approvedByEmployeeNo references Employee(employeeNo)
DocumentAuthor	(<u>documentNo</u> , <u>employeeNo</u>) Primary Key documentNo, employeeNo Foreign Key documentNo references Document(documentNo) Foreign Key employeeNo references Employee(employeeNo)
Employee	(<u>employeeNo</u> , dateStartRole, firstName, middleName, lastName, address, workTelExt, homeTelNo, empEmailAddress, socialSecurityNumber, DOB, position, sex, salary, dateStarted, roleNo) Primary Key employeeNo Alternate Key socialSecurityNumber Foreign Key roleNo references Role(roleNo)
Expense	(<u>expenseNo</u> , expenseDate, expenseDescription, expenseAmount, workPackageNo, employeeNo, expenseTypeNo) Primary Key expenseNo Alternate Key workPackageNo, employeeNo, expenseDate Foreign Key workPackageNo references WorkPackage(workPackageNo) Foreign Key employeeNo references Employee(employeeNo) Foreign Key expenseTypeNo reference ExpenseType(expenseTypeNo)
ExpenseType	(<u>expenseTypeNo</u> , expenseTypeDescription) Primary Key expenseTypeNo
Project	(<u>projectNo</u> , projectName, plannedStartDate, plannedEndDate, actualStartDate, actualEndDate, projectedCost, actualCost, clientNo, managerEmployeeNo) Primary Key projectNo Foreign Key clientNo references Client(clientNo) Foreign Key managerEmployeeNo references Employee(employeeNo)
Role	(<u>roleNo</u> , roleDescription, billingRate) Primary Key roleNo
TimeBooked	(<u>workPackageNo</u> , <u>employeeNo</u> , dateStartWork, dateStopWork, timeWorked) Primary Key workPackageNo, employeeNo Foreign Key workPackageNo references WorkPackage(workPackageNo) Foreign Key employeeNo references Employee(employeeNo)
WorkPackage	(<u>workPackageNo</u> , plannedStartDate, plannedEndDate, actualStartDate, actualEndDate, projectedCost, actualCost, projectNo) Primary Key workPackageNo Foreign Key projectNo references Project(projectNo)

E.5 Course management

A training company wishes to create a database of its course information. The company delivers a number of seminars and training courses. Each course is delivered by one member of staff at some location (such as internal seminar room S10, Hilton Hotel Suite 100). The fees vary for each course and on the number of delegates a company sends. For example, if a company sends one person, the charge may be \$1000. If the company sends two people, the first may be charged \$1000, but the second may be charged \$750. The course can be attended by a number of delegates, subject to some upper limit for the course. A delegate can register as an individual or through his or her company. The name of the employee who registers the delegate is recorded. An invoice is sent either to the delegate or to his or her company. The logical data model is shown in Figure E.9 and the associated tables in Figure E.10.

Figure E.9

Logical data model for course management.

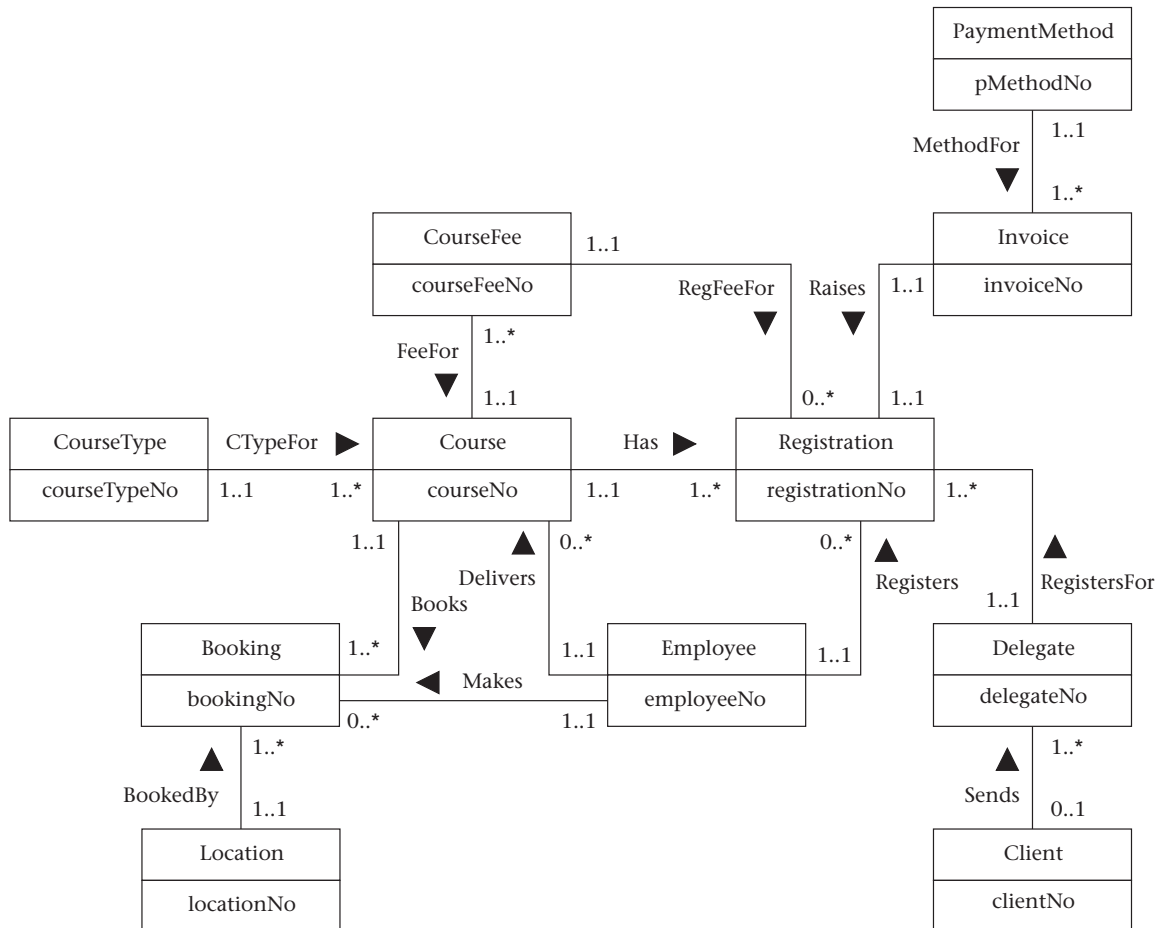


Figure E.10

Tables for course management.

Client	As defined in Section D.4.2
Employee	As defined in Section D.1.2
PaymentMethod	As defined in Section D.1.2
Delegate	(<u>delegateNo</u> , delegateTitle, delegateFName, delegateLName, delegateStreet, delegateCity, delegateState, delegateZipCode, attTelNo, attFaxNo, attEmailAddress, clientNo) Primary Key delegateNo Foreign Key clientNo references Client(clientNo)
Booking	(<u>bookingNo</u> , bookingDate, locationNo, courseNo, bookingEmployeeNo) Primary Key bookingNo Foreign Key locationNo references Location(locationNo) Foreign Key courseNo references Course(courseNo) Foreign Key bookingEmployeeNo references Employee(employeeNo)
Course	(<u>courseNo</u> , courseName, courseDescription, startDate, startTime, endDate, endTime, maxDelegates, confirmed, delivererEmployeeNo, courseTypeNo) Primary Key courseNo Foreign Key delivererEmployeeNo references Employee(employeeNo) Foreign Key courseTypeNo references CourseType(courseTypeNo)
CourseFee	(<u>courseFeeNo</u> , feeDescription, fee, courseNo) Primary Key courseFeeNo Foreign Key courseNo references Course(courseNo)
CourseType	(<u>courseTypeNo</u> , courseTypeDescription) Primary Key courseTypeNo
Invoice	(<u>invoiceNo</u> , dateRaised, datePaid, creditCardNo, holdersName, expiryDate, registrationNo, pMethodNo) Primary Key invoiceNo Foreign Key registrationNo references Registration(registrationNo) Foreign Key pMethodNo references PaymentMethod(pMethodNo)
Location	(<u>locationNo</u> , locationName, maxSize) Primary Key locationNo
Registration	(<u>registrationNo</u> , registrationDate, delegateNo, courseFeeNo, registerEmployeeNo, courseNo) Primary Key registrationNo Foreign Key delegateNo references Delegate(delegateNo) Foreign Key courseFeeNo references CourseFee(courseFeeNo) Foreign Key registerEmployeeNo references Employee(employeeNo) Foreign Key courseNo references Course(courseNo)

E.6 Human resource management

An HRM Department wishes to create a database to monitor its employees. The company is divided into a number of departments, and employees are assigned to one department. The department has a designated Manager who has overall responsibility for the department and the employees in the department. However, to help manage the department, a number of employees are nominated to supervise groups of staff. When a new employee joins the company, information on previous work history and qualifications is required. On a regular basis, each employee is required to undergo a review, which is normally carried out by the Manager, but may be delegated to a nominated representative.

The company has defined a number of position types, such as Manager, Business Analyst, Salesperson, Secretary, and each type has a number of grades associated with it, which for most non-senior positions determines the employee's salary. At a senior level, salary is negotiable. Posts are allocated to a department depending on its workload. For example, a department may be allocated two new Business Analyst posts. A post will be filled by one employee, although over time, employees will fill a number of different posts.

The logical data model is shown in Figure E.11 and the associated tables in Figure E.12.

Figure E.11

Logical data model for human resource management.

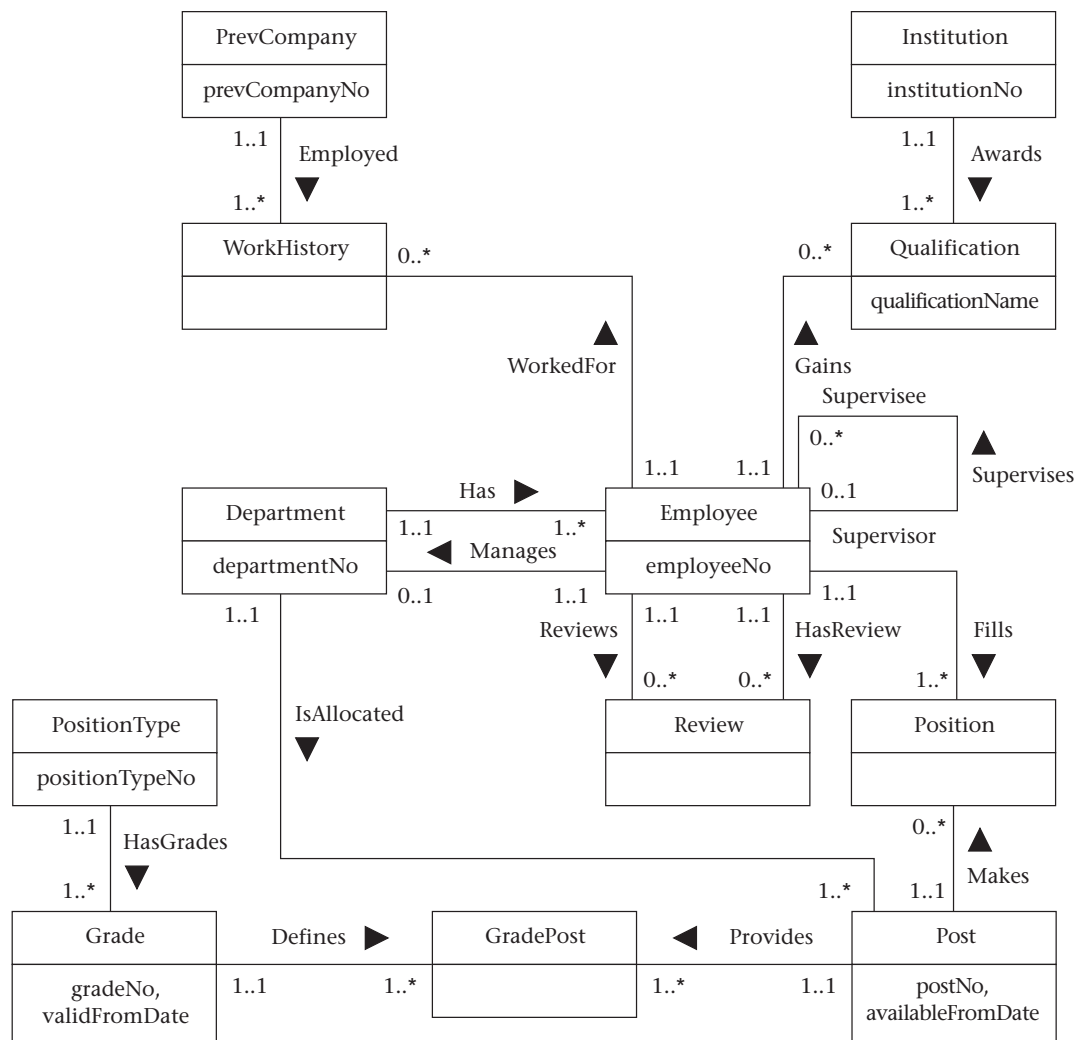


Figure E.12

Tables for human resource management.

Department	(<u>departmentNo</u>, departmentName, deptLocation, managerEmployeeNo) Primary Key departmentNo Foreign Key managerEmployeeNo references Employee(employeeNo)
Employee	(<u>employeeNo</u>, title, firstName, middleName, lastName, address, workTelExt, homeTelNo, empEmailAddress, socialSecurityNumber, DOB, position, sex, salary, dateStarted, dateLeft, departmentNo, supervisorEmployeeNo) Primary Key employeeNo Alternate Key socialSecurityNumber Foreign Key departmentNo references Department(departmentNo) Foreign Key supervisorEmployeeNo references Employee(employeeNo)
Grade	(<u>gradeNo</u>, <u>validFromDate</u>, validToDate, gradeDescription, gradeSalary, noDaysLeaveEntitlement, positionTypeNo) Primary Key gradeNo, validFromDate Foreign Key positionTypeNo references PositionType(positionTypeNo)
GradePost	(<u>gradeNo</u>, <u>validFromDate</u>, <u>postNo</u>, <u>availableFromDate</u>) Primary Key gradeNo, validFromDate, postNo, availableFromDate Foreign Key gradeNo, validFromDate references Grade(gradeNo, validFromDate) Foreign Key postNo, availableFromDate references Post(postNo, availableFromDate)
Institution	(<u>institutionNo</u>, institutionName, instAddress, instTelNo, instFaxNo, instWebAddress, contactName, contactTelNo, contactFaxNo, contactEmailAddress) Primary Key institutionNo Alternate Key institutionName Alternate Key instTelNo Alternate Key instFaxNo
Position	(<u>employeeNo</u>, <u>postNo</u>, <u>startDate</u>, endDate) Primary Key employeeNo, postNo, startDate Foreign Key employeeNo references Employee(employeeNo) Foreign Key postNo, startDate references Post(postNo, availableFromDate)
PositionType	(<u>positionTypeNo</u>, positionTypeDescription) Primary Key positionTypeNo
Post	(<u>postNo</u>, <u>availableFromDate</u>, availableToDate, postDescription, salariedHourly, fullPartTime, temporaryPermanent, freeLaborStandardsActExempt, departmentNo) Primary Key postNo, availableFromDate Foreign Key departmentNo references Department(departmentNo)
PrevCompany	(<u>prevCompanyNo</u>, pCompanyName, pCompanyStreet, pCompanyCity, pCompanyState, pCompanyZipCode, pCompanyTelNo, pCompanyFaxNo, pCompanyWebAddress, contactName, contactTelNo, contactFaxNo, contactEmailAddress) Primary Key prevCompanyNo Alternate Key pCompanyName Alternate Key pCompanyTelNo Alternate Key pCompanyFaxNo
Qualification	(<u>qualificationName</u>, <u>employeeNo</u>, gradeObtained, startQualDate, endQualDate, gpa, institutionNo) Primary Key qualificationName, employeeNo Foreign Key employeeNo references Employee(employeeNo) Foreign Key institutionNo references Institution(institutionNo)

Figure E.12

Continued

Review	(<u>revieweeEmployeeNo</u> , <u>reviewerEmployeeNo</u> , <u>reviewDate</u> , comments) Primary Key revieweeEmployeeNo, reviewerEmployeeNo, reviewDate Foreign Key revieweeEmployeeNo references Employee(employeeNo) Foreign Key reviewerEmployeeNo references Employee(employeeNo)
WorkHistory	(<u>prevCompanyNo</u> , <u>employeeNo</u> , prevPosition, prevGrade, prevSalary, prevLocation, prevResponsibilities) Primary Key prevCompanyNo, employeeNo Foreign Key prevCompanyNo references PrevCompany(prevCompanyNo) Foreign Key employeeNo references Employee(employeeNo)

E.7 Payroll management

The Payroll Department wishes to create a database to monitor employees' salary payments. To calculate an employee's salary, Payroll need to take into consideration holidays taken against holiday entitlement, number of days' sick leave in pay period, bonuses, and deductions. An employee must specify how his or her salary should be paid, although this may change over time. Most employees are paid by electronic bank transfer, but some types of employees may be paid by cash or check. If payment is electronic, then a routing number and account type are required. Payment can be made by only one method. There are various reasons for deductions being made; for example, federal tax, state tax, medical plan, retirement plan, or cash advance.

The logical data model is shown in Figure E.13 and the associated tables in Figure E.14.

Figure E.13

Logical data model for payroll management.

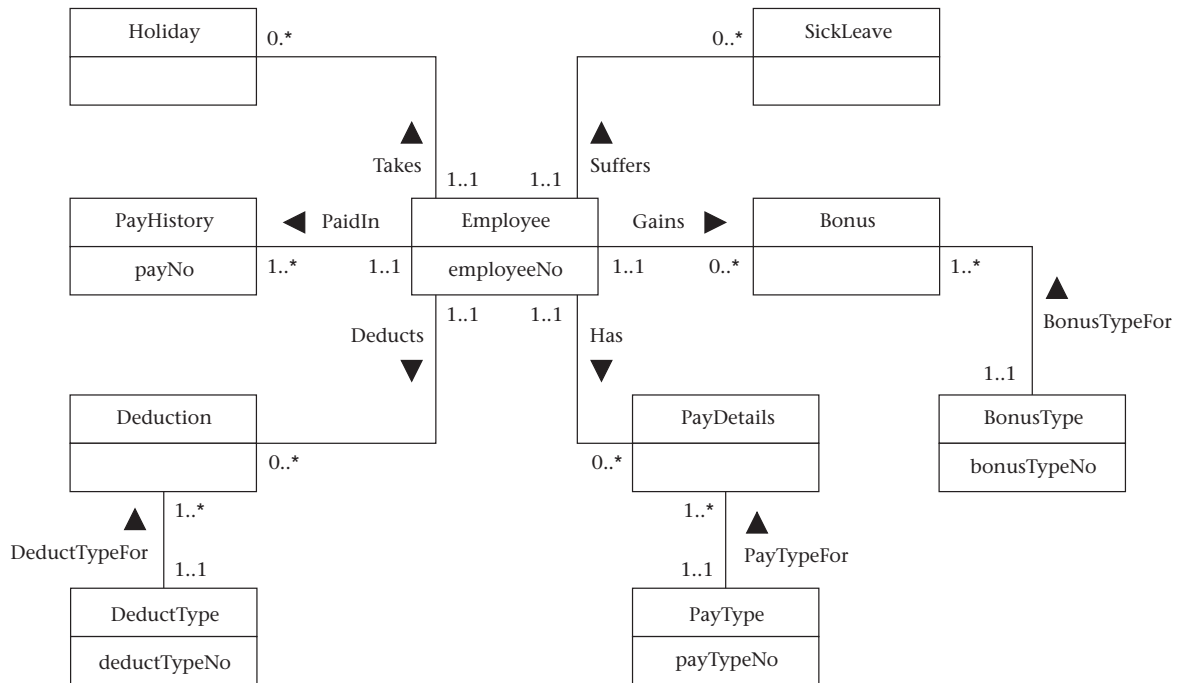


Figure E.14

Tables for payroll management.

Employee	As defined in Section D.1.2
Bonus	(<u>employeeNo</u> , <u>bonusDate</u> , bonusAmount, bonusTypeNo) Primary Key employeeNo, bonusDate Foreign Key employeeNo references Employee(employeeNo) Foreign Key bonusTypeNo references BonusType(bonusTypeNo)
BonusType	(<u>bonusTypeNo</u> , bonusDescription) Primary Key bonusTypeNo
Deduction	(<u>employeeNo</u> , <u>deductDate</u> , deductAmount, deductTypeNo) Primary Key employeeNo, deductDate Foreign Key employeeNo references Employee(employeeNo) Foreign Key deductTypeNo references DeductType(deductTypeNo)
DeductType	(<u>deductTypeNo</u> , deductDescription) Primary Key deductTypeNo
Holiday	(<u>employeeNo</u> , <u>startDate</u> , endDate) Primary Key employeeNo, startDate Foreign Key employeeNo references Employee(employeeNo)
PayDetails	(<u>employeeNo</u> , <u>startDate</u> , routingNumber, accountType, bankName, bankAddress, payTypeNo) Primary Key employeeNo, startDate Foreign Key employeeNo references Employee(employeeNo) Foreign Key payTypeNo references PayType(payTypeNo)
PayHistory	(<u>payNo</u> , employeeNo, payDate, checkNumber, payAmount) Primary Key payNo Foreign Key employeeNo references Employee(employeeNo)
PayType	(<u>payTypeNo</u> , payTypeDescription) Primary Key payTypeNo
SickLeave	(<u>employeeNo</u> , <u>startDate</u> , endDate, reason) Primary Key employeeNo, startDate Foreign Key employeeNo references Employee(employeeNo)

E.8 Vehicle rentals

A vehicle rental company wishes to create a database to monitor the renting of vehicles to clients. The company has various outlets and each outlet has staff including a Manager and several Senior Mechanics who are responsible for supervising the work of allocated groups of Mechanics. Each outlet has a stock of vehicles for rent that may be rented by clients for various periods of time, from a minimum of four hours to a maximum of six months. Each rental agreement between a client and the company is uniquely identified using a rental number. A client must take out insurance cover for each vehicle rental period. Each vehicle is checked for faults after each rental. The logical data model is shown in Figure E.15 and the associated tables in Figure E.16.

Figure E.15

Logical data model for vehicle rentals.

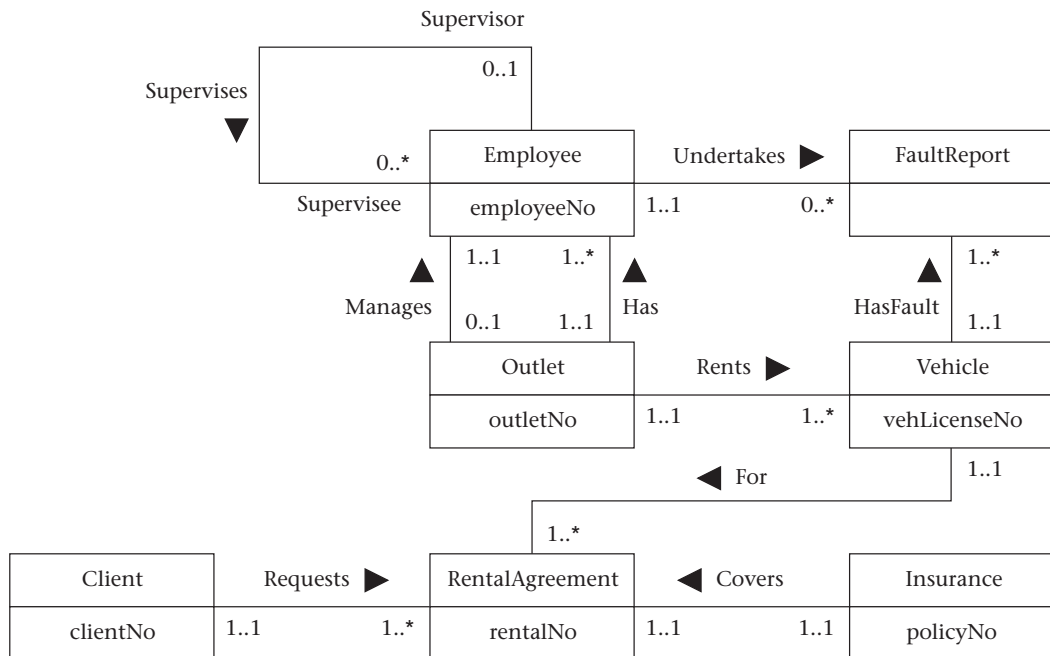


Figure E.16

Tables for vehicle rentals.

Client	As defined in Section D.4.2
Employee	(<u>employeeNo</u>, title, firstName, middleName, lastName, address, workTelExt, homeTelNo, empEmailAddress, socialSecurityNumber, DOB, position, sex, salary, dateStarted, outletNo, supervisorEmployeeNo) Primary Key employeeNo Alternate Key socialSecurityNumber Foreign Key outletNo references Outlet(outletNo) Foreign Key supervisorEmployeeNo references Employee(employeeNo)
FaultReport	(<u>vehLicenseNo</u>, <u>dateChecked</u>, timeChecked, comments, employeeNo) Primary Key vehLicenseNo, dateChecked Foreign Key vehLicenseNo references Vehicle(vehLicenseNo) Foreign Key employeeNo references Employee(employeeNo)
Outlet	(<u>outletNo</u>, outletStreet, outletCity, outletState, outletZipCode, outletTelNo, outletFaxNo, managerEmployeeNo) Primary Key outletNo Alternate Key outletTelNo Alternate Key outletFaxNo Foreign Key managerEmployeeNo references Employee(employeeNo)
RentalAgreement	(<u>rentalNo</u>, dateStart, timeStart, dateReturn, timeReturn, mileageBefore, mileageAfter, policyNo, insuranceCoverType, insurancePremium, clientNo, vehLicenseNo) Primary Key rentalNo Alternate Key policyNo Foreign Key clientNo references Client(clientNo) Foreign Key vehLicenseNo references Vehicle(vehLicenseNo)
Vehicle	(<u>vehLicenseNo</u>, vehicleMake, vehicleModel, color, noDoors, capacity, hireRate, outletNo) Primary Key vehLicenseNo Foreign Key outletNo references Outlet(outletNo)

E.9 Student accommodation

The Accommodation Office of a university wishes to create a database to monitor the allocation of accommodation to students. Each student requiring accommodation fills out an application form, which holds the student's details and an indication of the type of accommodation required and the duration. Students may rent a room in a hall of residence or student apartment. The halls provide only single rooms, which have a room number, place number, and monthly rental rate. The place number uniquely identifies each room in all the halls controlled by the Accommodation Office and is used when renting a room to a student. Each hall is managed by a member of the Accommodation Office.

The Accommodation Office also offers student apartments, each identified by a unique apartment number. These apartments are fully furnished and provide single room accommodation for groups of three, four, or five students. Each bedroom in an apartment has a monthly rental rate, a room number, and a place number. The place number uniquely identifies each room available in all student apartments and is used when renting a room to a student. Apartments are inspected by members of the Accommodation Office on a regular basis to ensure that the accommodation is well maintained.

New lease agreements are negotiated at the start of each academic year with a minimum rental period of one semester and a maximum rental period of one year. The students pay for their accommodation throughout the academic year and are sent an invoice at the start of each semester. If a student does not pay by a certain date, two reminder letters are sent. The logical data model is shown in Figure E.17 and the associated tables in Figure E.18.

Figure E.17

Logical data model for student accommodation.

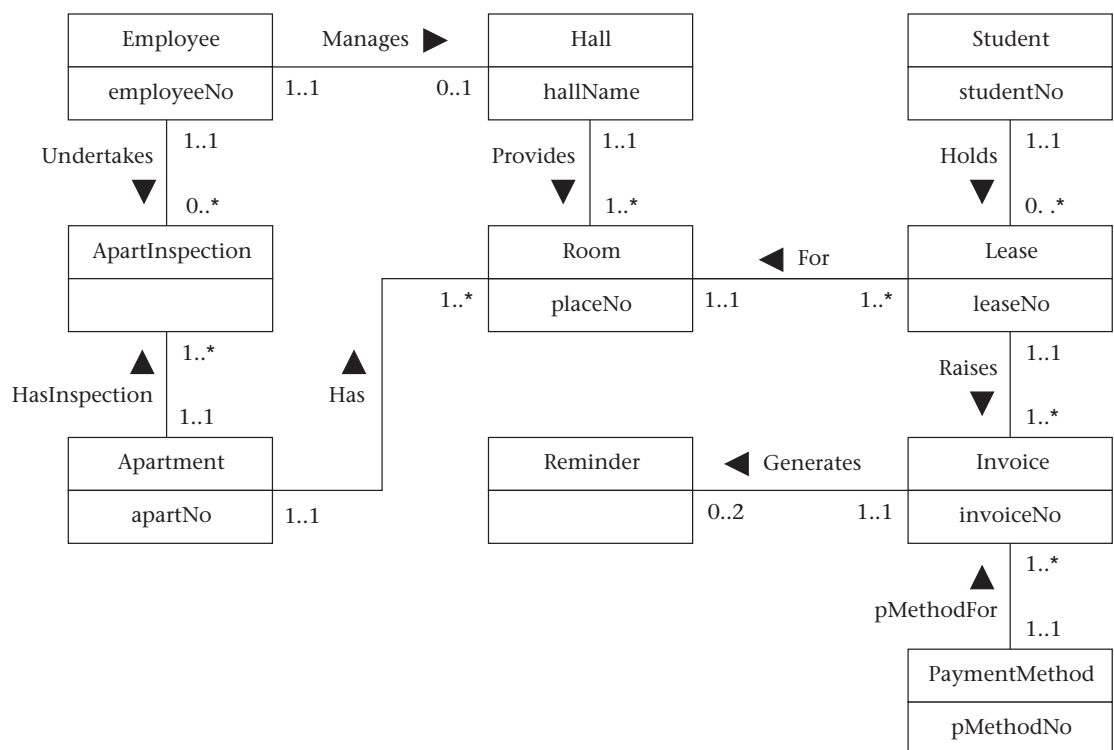


Figure E.18

Tables for student accommodation.

Employee	As defined in Section D.1.2
PaymentMethod	As defined in Section D.1.2
Apartment	(<u>apartNo</u> , apartAddress, noOfRoomsInApart) Primary Key apartNo
ApartmentInspection	(<u>apartNo</u> , <u>dateOfInspection</u> , comments, status, employeeNo) Primary Key apartNo, dateOfInspection Foreign Key apartNo references Apartment(apartNo) Foreign Key employeeNo references Employee(employeeNo)
Hall	(<u>hallName</u> , hallAddress, hallTelNo, hallFaxNo, noOfRoomsInHall, managerEmployeeNo) Primary Key hallName Alternate Key hallTelNo Alternate Key hallFaxNo Foreign Key managerEmployeeNo references Employee(employeeNo)
Invoice	(<u>invoiceNo</u> , semester, dateDue, datePaid, leaseNo, pMethodNo) Primary Key invoiceNo Foreign Key leaseNo references Lease(leaseNo) Foreign Key pMethodNo references PaymentMethod(pMethodNo)
Reminder	(<u>invoiceNo</u> , dateReminder1sent, dateReminder2sent, dateInterview, comments) Primary Key invoiceNo Foreign Key invoiceNo references Invoice(invoiceNo)
Lease	(<u>leaseNo</u> , duration, dateStart, dateLeave, studentNo, placeNo) Primary Key leaseNo Alternate Key placeNo, dateStart Alternate Key studentNo, dateStart Foreign Key studentNo references Student(studentNo) Foreign Key placeNo references Room(placeNo)
Room	(<u>placeNo</u> , roomNo, rentPerSemester, hallName, apartNo) Primary Key placeNo Alternate Key roomNo, hallName Alternate Key roomNo, apartNo Foreign Key hallName references Hall(hallName) Foreign Key apartNo references Apartment(apartNo)
Student	(<u>studentNo</u> , studentFirstName, studentMiddleInitial, studentLastName, studentHomeStreet, studentHomeCity, studentHomeState, studentHomeZipCode, studentHomeTelNo, studentSex, studentDOB, studentType, studentStatus, accommodationTypeRequired, accommodationDuration) Primary Key studentNo

E.10 Client transportation

A haulage company that specializes in the transportation of loads throughout the US wishes to create a database to control client orders for transportation. The company has many offices throughout the US to process client orders. A client registers with an office and can place one or more orders. Each order describes the load to be transported along with the collection address and the delivery address. The transportation requirements for each order are then calculated. The transport requirements describe the number of units and trailers required to transport the load. Each office is allocated several units and trailers. One unit can pull one or two trailers. The logical data model is shown in Figure E.19 and the associated tables in Figure E.20.

Figure E.19

Logical data model for client transportation.

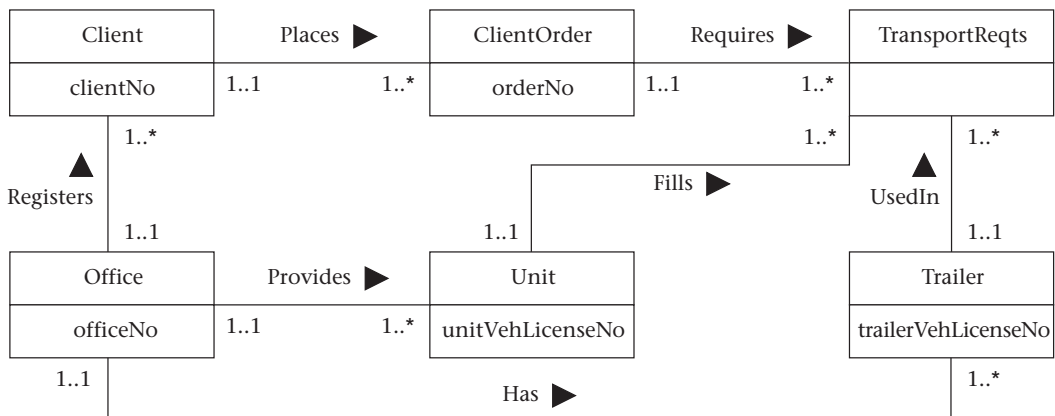


Figure E.20

Tables for client transportation.

Client	(<u>clientNo</u>, clientName, clientStreet, clientCity, clientState, clientZipCode, clientTelNo, clientFaxNo, clientWebAddress, contactName, contactTelNo, contactFaxNo, contactEmailAddress, officeNo) Primary Key clientNo Alternate Key clientTelNo Alternate Key clientFaxNo Foreign Key officeNo references Office(officeNo)
Office	(<u>officeNo</u>, officeAddress, officeTelNo, officeFaxNo) Primary Key officeNo Alternate Key officeTelNo Alternate Key officeFaxNo
ClientOrder	(<u>orderNo</u>, dateOrder, collectionDate, collectionAddress, deliveryDate, deliveryAddress, loadWeight, loadDescription, clientNo) Primary Key orderNo Foreign Key clientNo references Client(clientNo)
Trailer	(<u>trailerVehLicenseNo</u>, trailerDescription, trailerLength, maxCarryingWeight, officeNo) Primary Key trailerVehLicenseNo Foreign Key officeNo references Office(officeNo)
TransportReqs	(<u>orderNo</u>, <u>transportReqPartNo</u>, unitVehLicenseNo, trailerVehLicenseNo1, trailerVehLicenseNo2) Primary Key orderNo, transportReqPartNo Foreign Key unitVehLicenseNo references Unit(unitVehLicenseNo) Foreign Key trailerVehLicenseNo1 references Trailer(trailerVehLicenseNo) Foreign Key trailerVehLicenseNo2 references Trailer(trailerVehLicenseNo)
Unit	(<u>unitVehLicenseNo</u>, unitDescription, maxPayLoad, officeNo) Primary Key unitVehLicenseNo Foreign Key officeNo references Office(officeNo)

E.11 Publisher printing

A printing company that handles print jobs for book publishers wishes to create a database to control client requests for printing. A book publisher submits an order that describes the printing job. A print job requires the use of materials, such as paper and ink, which are assigned to a job through one or more purchase orders. Each print job is assigned to a Print Manager, who has the responsibility to ensure that the job is carried out correctly. For larger print jobs, additional employees are normally allocated to help with the printing. The logical data model is shown in Figure E.21 and the associated tables in Figure E.22.

Figure E.21

Logical data model for publisher printing.

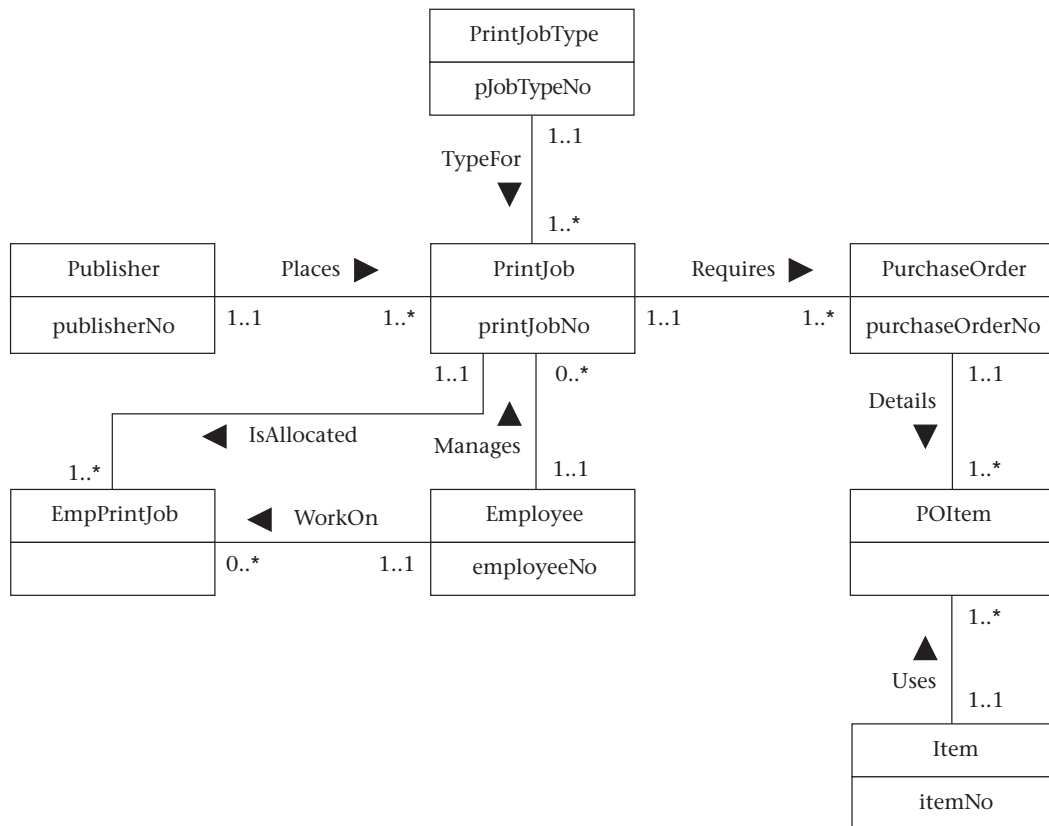


Figure E.22

Tables for publisher printing.

Employee	As defined in Section D.1.2
EmpPrintJob	(<u>employeeNo</u> , <u>printJobNo</u> , jobDate) Primary Key employeeNo, printJobNo Foreign Key employeeNo references Employee(employeeNo) Foreign Key printJobNo references PrintJob(printJobNo)
Item	(<u>itemNo</u> , itemDescription, itemPrice, itemQuantityInStock, itemReorderLevel, itemReorderQuantity, itemReorderLeadTime) Primary Key itemNo
PrintJob	(<u>printJobNo</u> , printJobDescription, printJobDateReceived, printJobDateCompleted, managerEmployeeNo, publisherNo, printJobTypeNo) Primary Key printJobNo Foreign Key managerEmployeeNo references Employee(employeeNo) Foreign Key publisherNo references Publisher(publisherNo) Foreign Key printJobTypeNo references PrintJobType(printJobTypeNo)
Publisher	(<u>publisherNo</u> , publisherName, publisherStreet, publisherCity, publisherState, publisherZipCode, pubTelNo, pubFaxNo, pubWebAddress, contactName, contactTelNo, contactFaxNo, contactEmailAddress, creditRating) Primary Key publisherNo Alternate Key publisherName Alternate Key pubTelNo Alternate Key pubFaxNo
POItem	(<u>purchaseOrderNo</u> , <u>itemNo</u> , quantity) Primary Key purchaseOrderNo, itemNo Foreign Key purchaseOrderNo references PurchaseOrder (purchaseOrderNo) Foreign Key itemNo references Item(itemNo)
PrintJobType	(<u>printJobTypeNo</u> , printJobTypeDescription) Primary key printJobTypeNo
PurchaseOrder	(<u>purchaseOrderNo</u> , <u>printJobNo</u> , purchaseOrderDate) Primary Key purchaseOrderNo Foreign Key printJobNo references PrintJob(printJobNo)

E.12 County library

A county wishes to create a database to control its local libraries. Each library has a number of employees, one of whom is designated as the manager of the library and is responsible for supervising employees and the general day-to-day management of the library. Each library stores a number of books and CDs. A citizen has to become a member of a library before he or she is allowed to borrow any books, but thereafter can borrow books from any county library. Books are stored on shelves and CDs are stored in a number of racks in the center of the library. Generally, a library stocks a number of copies of each book title and each CD. Details of book publishers are maintained but not CD publishers. To find an item, searches can be performed based on the book/CD title, the author/artist's name, the category of the book/CD, or the publisher's name. The logical data model is shown in Figure E.23 and the associated tables in Figure E.24.

Figure E.23

Logical data model for county library.

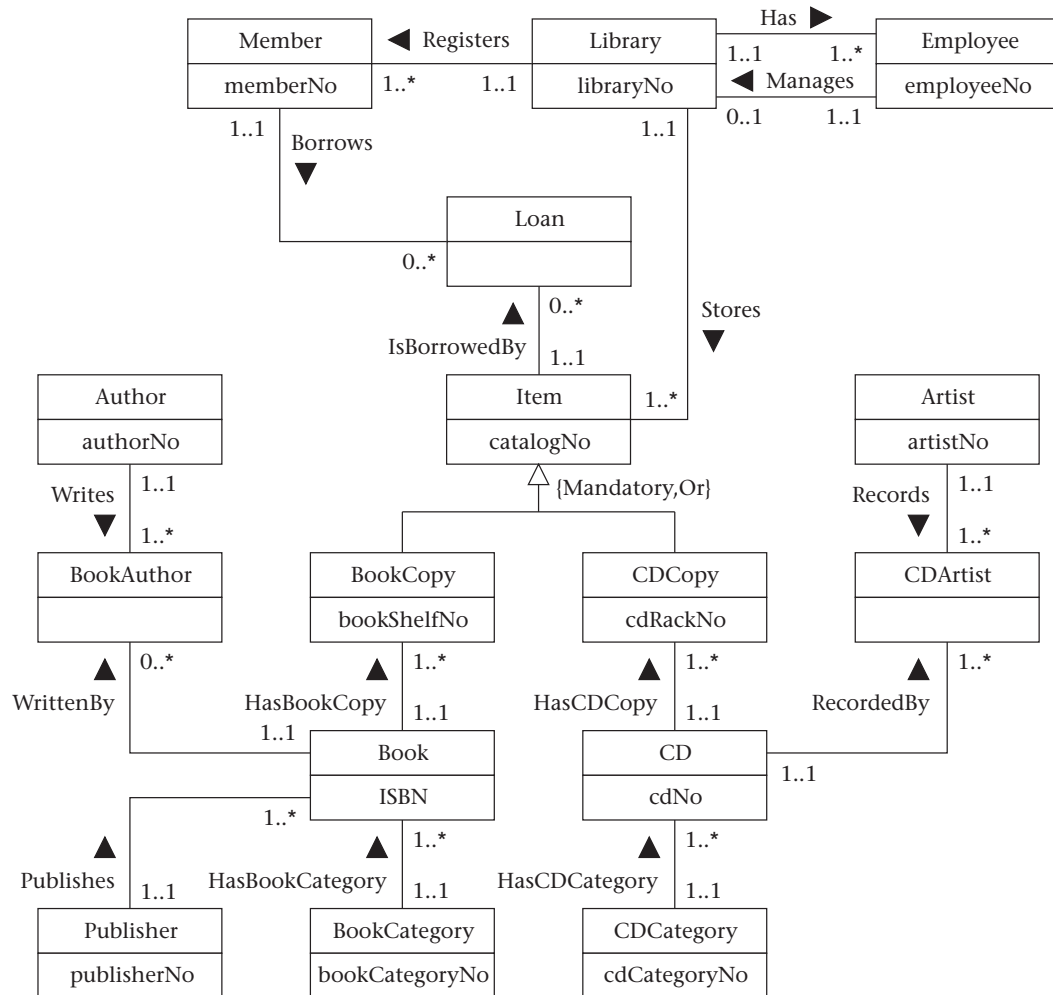


Figure E.24

Tables for county library.

Publisher	As defined in Section D.11.2
Artist	(<u>artistNo</u> , name) Primary Key artistNo
Author	(<u>authorNo</u> , name) Primary Key authorNo
Book	(<u>ISBN</u> , title, year, publisherNo, bookCategoryNo) Primary Key ISBN Foreign Key publisherNo references Publisher(publisherNo) Foreign Key bookCategoryNo references BookCategory(bookCategoryNo)
BookAuthor	(<u>ISBN</u> , <u>authorNo</u>) Primary Key ISBN, authorNo Foreign Key ISBN references Book(ISBN) Foreign Key authorNo references Author(authorNo)
BookCategory	(<u>bookCategoryNo</u> , bookCatDescription) Primary Key bookCategoryNo
BookCopy	(<u>catalogNo</u> , bookShelfNo, ISBN, dateInStock, libraryNo) Primary Key catalogNo Alternate Key bookShelfNo Foreign Key ISBN references Book(ISBN) Foreign Key libraryNo references Library(libraryNo)
CD	(<u>cdNo</u> , title, releaseDate, cdCategoryNo) Primary Key cdNo Foreign Key cdCategoryNo references CDCategory(cdCategoryNo)
CDArtist	(<u>cdNo</u> , <u>artistNo</u>) Primary Key cdNo, artistNo Foreign Key cdNo references CD(cdNo) Foreign Key artistNo references Artist(artistNo)
CDCategory	(<u>cdCategoryNo</u> , cdCatDescription) Primary Key cdCategoryNo
CDCopy	(<u>catalogNo</u> , cdRackNo, cdNo, dateInStock, libraryNo) Primary Key catalogNo Alternate Key cdRackNo Foreign Key cdNo references CD(cdNo) Foreign Key libraryNo references Library(libraryNo)
Employee	(<u>employeeNo</u> , title, firstName, middleName, lastName, address, workTelExt, homeTelNo, empEmailAddress, socialSecurityNumber, DOB, position, sex, salary, dateStarted, libraryNo) Primary Key employeeNo Alternate Key socialSecurityNumber Foreign Key libraryNo references Library(libraryNo)
Library	(<u>libraryNo</u> , libStreet, libCity, libState, libZipCode, libTelNo, libFaxNo, libWebAddress, managerEmployeeNo) Primary Key libraryNo Alternate Key libTelNo Alternate Key libFaxNo

Figure E.24

Continued

	Foreign Key managerEmployeeNo references Employee(employeeNo)
Loan	(<u>catalogNo</u> , <u>memberNo</u> , dateOut, dateReturn) Primary Key catalogNo, memberNo Foreign Key catalogNo references BookCopy(catalogNo) and CDCopy(catalogNo) Foreign Key memberNo references Member(memberNo)
Member	(<u>memberNo</u> , memTitle, memFirstName, memMiddleName, memLastName, memAddress, memWorkTelExt, memHomeTelNo, memDOB, memSex, dateJoined, libraryNo) Primary Key memberNo Foreign Key libraryNo references Library(libraryNo)

E.13 Real estate rentals

A real estate agency with branches throughout the US wishes to create a database to control the properties it rents out on behalf of owners, who are classified as Business and Private owners. Within each branch, staff oversee the rental of properties, and are responsible for handling property viewings and lease agreements. Some staff are also given the role Supervisor, responsible for overseeing a group of staff and ensuring the efficient management of the branch. The administrative work of each group of staff is supported by a Secretary. The logical data model is shown in Figure E.25 and the associated tables in Figure E.26.

Figure E.25

Logical data model for real estate agency.

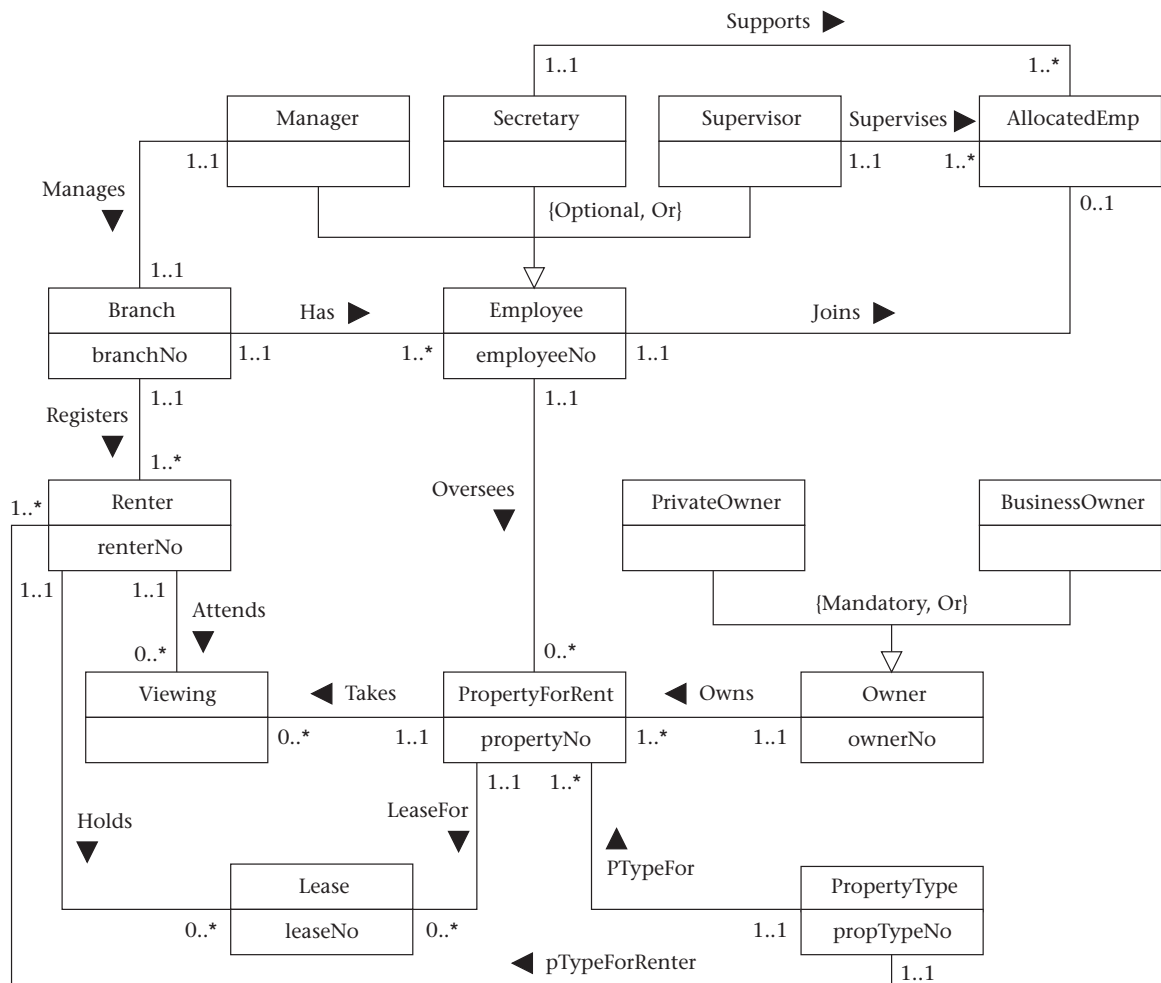


Figure E.26

Tables for real estate agency.

AllocatedEmp	(<u>superviseeEmployeeNo</u>, supervisorEmployeeNo, secretaryEmployeeNo) Primary Key superviseeEmployeeNo Foreign Key superviseeEmployeeNo references Employee(employeeNo) Foreign Key supervisorEmployeeNo references Employee(employeeNo) Foreign Key secretaryEmployeeNo references Employee(employeeNo)
Branch	(<u>branchNo</u>, branchStreet, branchCity, branchState, branchZipCode, branchTelNo, branchFaxNo, managerEmployeeNo) Primary Key branchNo Alternate Key branchTelNo Alternate Key branchFaxNo Foreign Key managerEmployeeNo references Employee(employeeNo)
BusinessOwner	(<u>ownerNo</u>, businessName, businessAddress, businessTelNo, businessFaxNo, contactName, contactTelNo, contactFaxNo, contactEmailAddress) Primary Key ownerNo Alternate Key businessName Alternate Key businessTelNo Alternate Key businessFaxNo
Employee	(<u>employeeNo</u>, title, firstName, middleName, lastName, address, workTelExt, homeTelNo, empEmailAddress, socialSecurityNumber, DOB, position, sex, salary, typingSpeed, dateStarted, branchNo) Primary Key employeeNo Alternate Key socialSecurityNumber Foreign Key branchNo references Branch(branchNo)
Lease	(<u>leaseNo</u>, rentStart, rentFinish, depositPaid, renterNo, propertyNo) Primary Key leaseNo Foreign Key renterNo references Renter(renterNo) Foreign Key propertyNo references PropertyForRent(propertyNo)
PrivateOwner	(<u>ownerNo</u>, ownerName, ownerAddress, ownerTelNo) Primary Key ownerNo
PropertyForRent	(<u>propertyNo</u>, propStreet, propCity, propState, propZipCode, noRooms, rent, propTypeNo, ownerNo, employeeNo, branchNo) Primary Key propertyNo Foreign Key propTypeNo references PropertyType(propTypeNo) Foreign Key ownerNo references PrivateOwner(ownerNo) and BusinessOwner(ownerNo) Foreign Key employeeNo references Employee(employeeNo) Foreign Key branchNo references Branch(branchNo)
PropertyType	(<u>propTypeNo</u>, propTypeDescription) Primary Key propTypeNo
Renter	(<u>renterNo</u>, rFName, rLName, rAddress, rTelNo, maxRent, prefTypeNo) Primary Key renterNo Foreign Key prefTypeNo references PropertyType(propTypeNo)
Viewing	(<u>propertyNo</u>, <u>renterNo</u>, <u>dateView</u>, comments) Primary Key propertyNo, renterNo, dateView Foreign Key propertyNo references PropertyForRent(propertyNo) Foreign Key renterNo references Renter(renterNo)

E.14 Travel agent

A travel agent wishes to create a database for its customer holiday booking activities. The travel agent has numerous branches spread through the major cities in the US. A customer can call into, or contact by telephone, any branch and book a holiday. A holiday normally includes a flight and accommodation, although sometimes customers require only a flight or only accommodation. Once the travel agent has found a suitable holiday for a customer, the flight and accommodation requirements are reserved for the customer. However, the reservation can only be held for up to 24 hours by which time the customer must accept or decline the booking. Once the booking is accepted, the customer is invoiced for the holiday and must pay the invoice in full, a minimum of four weeks before the departure date. The name of the employee who initially processes the customer booking is recorded. The ER model is shown in Figure E.27 and the associated tables in Figure E.28.

Figure E.27

ER model for travel agent.

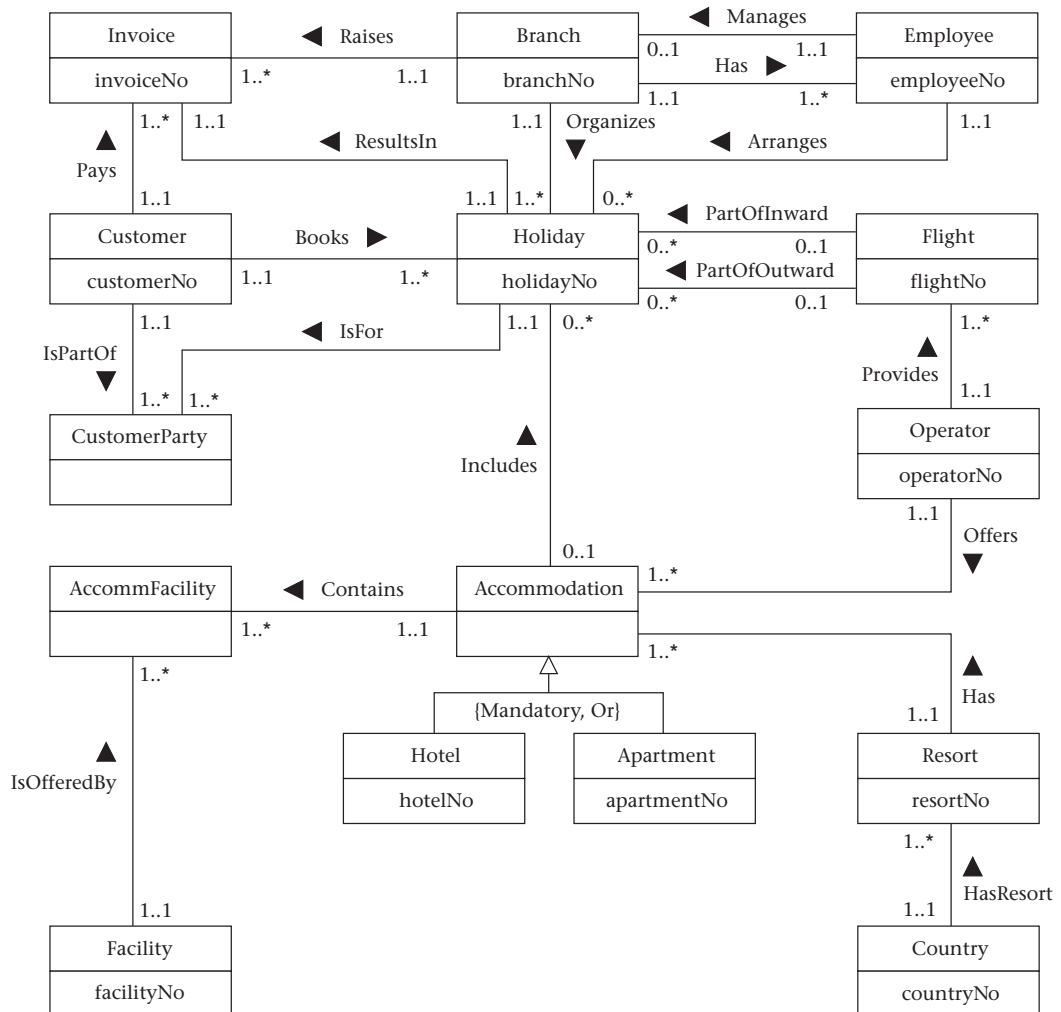


Figure E.28

Tables for travel agent.

Branch	As defined in Section D.13.2
ApartFacility	(<u>apartmentNo</u> , <u>facilityNo</u> , comments) Primary Key apartmentNo, facilityNo Foreign Key apartmentNo references Apartment(apartmentNo) Foreign Key facilityNo references Facility(facilityNo)
Apartment	(<u>apartmentNo</u> , apartmentName, apartmentType, apartmentDescription, apartmentRating, apartmentStreet, apartmentCity, apartmentState, apartmentCountry, apartmentZipCode, noOfRooms, operatorNo, resortNo) Primary Key apartmentNo Foreign Key operatorNo references Operator(operatorNo) Foreign Key resortNo references Resort(resortNo)
Country	(<u>countryNo</u> , countryName) Primary Key countryNo Alternate Key countryName
Customer	(<u>customerNo</u> , customerName, customerStreet, customerCity, customerState, customerZipCode, custTelNo, custFaxNo, nationality, sex, DOB, passportNo) Primary Key customerNo Alternate Key custTelNo Alternate Key custFaxNo Alternate Key passportNo
CustomerParty	(<u>customerNo</u> , <u>holidayNo</u>) Primary Key customerNo, holidayNo Foreign Key customerNo references Customer(customerNo) Foreign Key holidayNo references Holiday(holidayNo)
Employee	(<u>employeeNo</u> , title, firstName, middleName, lastName, address, workTelExt, homeTelNo, empEmailAddress, socialSecurityNumber, DOB, position, sex, salary, dateStarted, branchNo) Primary Key employeeNo Alternate Key socialSecurityNumber Foreign Key branchNo references Branch(branchNo)
Facility	(<u>facilityNo</u> , description, additionalCharge) Primary Key facilityNo
Flight	(<u>flightNo</u> , planeType, seatCapacity, airportDepart, departTime, airportArrive, arriveTime, operatorNo) Primary Key flightNo Foreign Key operatorNo references Operator(operatorNo)
Hotel	(<u>hotelNo</u> , hotelName, hotelStreet, hotelCity, hotelState, hotelCountry, hotelZipCode, hotelTelNo, hotelFaxNo, hotelType, hotelDescription, hotelRating, hotelManagerName, operatorNo, resortNo) Primary Key hotelNo Foreign Key operatorNo references Operator(operatorNo) Foreign Key resortNo references Resort(resortNo)
HotelFacility	(<u>hotelNo</u> , <u>facilityNo</u> , comments) Primary Key hotelNo, facilityNo Foreign Key hotelNo references Hotel(hotelNo) Foreign Key facilityNo references Facility(facilityNo)

Figure E.28*Continued*

Holiday	(<u>holidayNo</u>, status, dateBooked, cateringType, startDate, finishDate, invoiceNo, totalCost, dateSent, datePaid, bookCustomerNo, hotelNo, apartmentNo, inwardFlightNo, inwardNoOfSeats, outwardFlightNo, outwardNoOfSeats, employeeNo, branchNo) Primary Key holidayNo Foreign Key bookCustomerNo references Customer(customerNo) Foreign Key hotelNo references Hotel(hotelNo) Foreign Key apartmentNo references Apartment(apartmentNo) Foreign Key inwardFlightNo references Flight(flightNo) Foreign Key outwardFlightNo references Flight(flightNo) Foreign Key employeeNo references Employee(employeeNo) ForeignKey branchNo references Branch(branchNo)
Operator	(<u>operatorNo</u>, operatorName, operatorType, operatorStreet, operatorCity, operatorState, operatorZipCode, operTelNo, operFaxNo, contactName, contactTelNo, contactFaxNo, contactEmailAddress) Primary Key operatorNo Alternate Key operTelNo Alternate Key operFaxNo
Resort	(<u>resortNo</u>, resortName, distanceFromAirport, timeFromAirport, countryNo) Primary Key resortNo Foreign Key countryNo references Country(countryNo)

E.15 Student results

A university wishes to create a database for recording the results of students. When a student joins the university he or she registers with a particular course. Each student is also assigned an Advisor of Studies. Each year of each course is made up of modules. The minimum and maximum number of modules that make up a year of a course is 6 and 8, respectively. A student must take and pass each module in a given year before he or she is allowed to move into the next year of the course or to graduate. A student is normally allowed three attempts to pass a module; however, additional attempts are allowed at the discretion of the university. A particular module can be offered as part of one or more courses.

The university has several departments, each of which offers a portfolio of courses. Each department has a Head of Department (HOD) and each course has a Course Leader. Each module is assigned to a member of staff called a Module Coordinator, who has the responsibility to oversee the teaching and the assessment of the module. The ER model is shown in Figure E.29 and the associated tables in Figure E.30.

Figure E.29

ER model for student results.

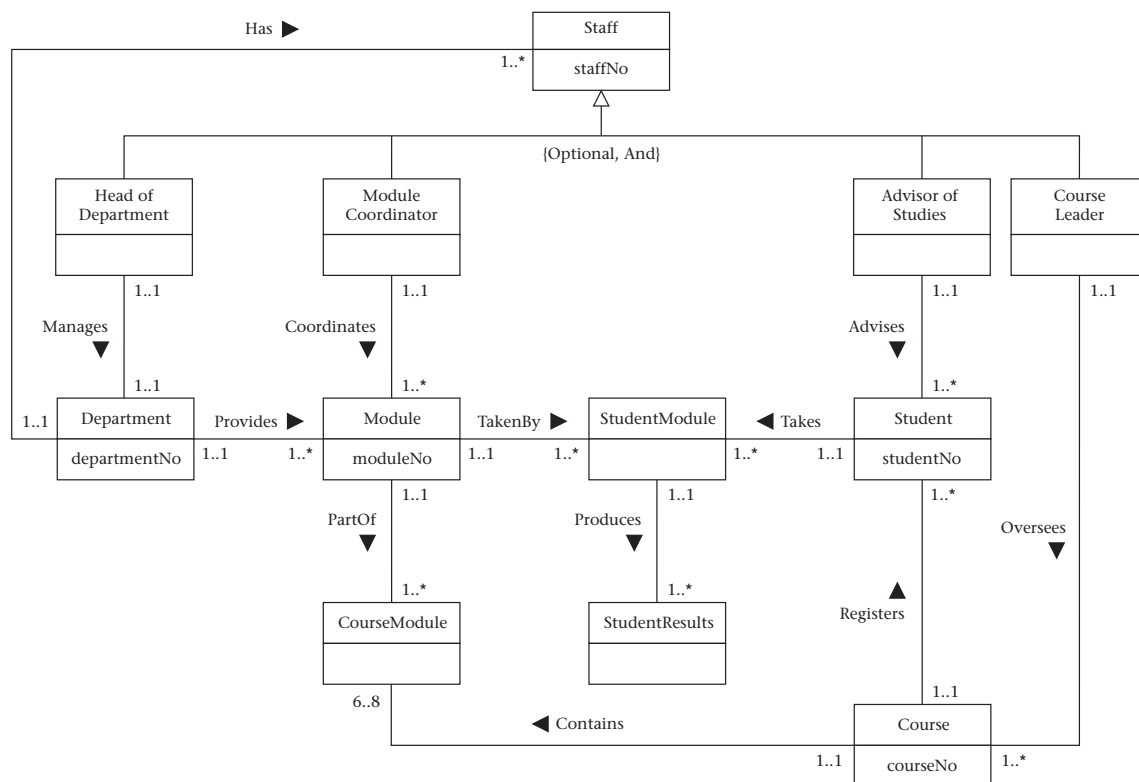


Figure E.30

Tables for student results.

Course	(<u>courseNo</u> , courseName, level, entranceRequirements, maxNumber, courseLeaderNo) Primary Key courseNo Alternate Key courseName Foreign Key courseLeaderNo references Staff(staffNo)
CourseModule	(<u>courseNo</u> , <u>moduleNo</u>) Primary Key courseNo, moduleNo Foreign Key courseNo references Course(courseNo) Foreign Key moduleNo references Module(moduleNo)
Department	(<u>departmentNo</u> , departmentName, location, HODstaffNo) Primary Key departmentNo Alternate Key departmentName Foreign Key HODstaffNo references Staff(staffNo)
Module	(<u>moduleNo</u> , moduleName, semesterDelivered, moduleAims, moduleObjectives, moduleSyllabus, moduleResources, moduleModeOfAssessment, moduleCoordinatorStaffNo, departmentNo) Primary Key moduleNo Alternate Key moduleName Foreign Key moduleCoordinatorStaffNo references Staff(staffNo) Foreign Key departmentNo references Department(departmentNo)
Staff	(<u>staffNo</u> , title, firstName, lastName, address, homeTelNo, workTelExt, empEmailAddress, socialSecurityNumber, DOB, position, sex, salary, dateStarted, departmentNo) Primary Key staffNo Alternate Key socialSecurityNumber Foreign Key departmentNo references Department(departmentNo)
Student	(<u>studentNo</u> , studentFirstName, studentMiddleName, studentLastName, studentHomeStreet, studentHomeCity, studentHomeState, studentHomeZipCode, studentHomeTelNo, familyHomeStreet, familyHomeCity, familyHomeState, familyHomeZipCode, familyHomeTelNo, studentDOB, studentSex, nationality, courseNo, advisorStaffNo) Primary Key studentNo Foreign Key courseNo references Course(courseNo) Foreign Key advisorStaffNo references Staff(staffNo)
StudentModule	(<u>studentNo</u> , <u>moduleNo</u>) Primary key studentNo, moduleNo Foreign Key studentNo references Student(studentNo) Foreign Key moduleNo references Module(moduleNo)
StudentResult	(<u>studentNo</u> , <u>moduleNo</u> , <u>attempt</u> , attemptDate, mark, proposal, additionalComments) Primary key studentNo, moduleNo, attempt Foreign Key studentNo, moduleNo references StudentModule(studentNo, moduleNo)